

PROBLEM SET 2

(due Jan 27, 2023 @ 8:30 AM in Gradescope and Moodle)

This problem set is to be completed **individually** and **submitted to both Gradescope and Moodle**. Problem 1 needs to be finished in Excel, and submitted to Moodle. The screenshots of Problem 1 excel results need to be included in the Gradescope submission. Problem 2-3 need to be finished in hardcopy format, and submitted to Gradescope.

Problem 1 (30 pts total)

The market values of townhomes are dependent on many factors, such as the floor space in square feet. A recent survey of a newly built townhome community shows the following data.

Space (sqft)	Value (\$)
2310	142,000
2333	144,000
2356	151,000
2379	150,000
2402	139,000
2425	169,000
2448	126,000
2471	142,900
2494	163,000

Copy the data into a blank worksheet and complete the following actions:

- Fitting the data by using the LINEST function. Obtain the slope and intercept of the linear fitting curve.
- Assume a linear correlation between the space area and asset value again. In the spreadsheet, now use Solver to determine the slope and intercept of the linear fit.

Please upload the excel spreadsheet to the **Moodle**, and attached the screenshots of the spreadsheet to the **Gradescope** submission.

Problem 2 (30 pts total)

How would you plot the following functions to make them look linear? What are the slope and the intercept (e.g., in the equation of a line, $y = m \cdot x + b$, y is the dependent variable and x is the independent variable; m and b are the slope and the intercept, respectively)?

- a) $q = 75 \log(x) - 3.5$ $x \dots$ independent variable, $q \dots$ dependent variable
- b) $t^{-1} = 11 + d$ $d \dots$ independent variable, $t \dots$ dependent variable
- c) $r = 1.123u - \frac{5}{3}$ $u \dots$ independent variable, $r \dots$ dependent variable
- d) $m = 4.18x^n$ $x \dots$ independent variable, $m \dots$ dependent variable, $n \dots$ constant.
- e) $m = 4.18x^n$ $n \dots$ independent variable, $m \dots$ dependent variable, $x \dots$ constant.

Problem 3 (40 pts total)

Enzymes (biological catalysts) lower the activation energy of a chemical reaction and thus increase the reaction rate, sometimes by a few orders of magnitude. An expression which describes enzyme activity is the Michaelis-Menten relationship:

$$v_0 = \frac{v_{\max} [S]}{K_m + [S]} \quad (1)$$

In Equation (1), v_0 is initial reaction rate (*i.e.*, change of concentration with time), $v_{\max}$ is maximum reaction rate (a constant value), K_m is the Michaelis-Menten constant, and $[S]$ is substrate concentration. The constants $v_{\max}$ and K_m can be determined from regression of data for the dependent variable v_0 as a function of the independent variable $[S]$.

- a) Write down the linearized form of Equation (1)
- b) What is the ordinate (y axis) when plotting the linear trend?
- c) What is the abscissa (x axis) when plotting the linear trend?
- d) Using the variables included in Equation (1), what is the expression for the slope of the linearized function?
- e) Using the variables included in Equation (1), what is the expression for the intercept of the linearized function?
- f) The initial reaction rate (v_0) is measured for two different substrate concentrations ($[S]$), with the resulting values: [10 mM/s, 0.5 mM] and [20 mM/s, 10 mM]. Based on these values, determine the corresponding values for $v_{\max}$ and K_m .

